

Refresher: Functions, Graphs and Equations

Rail Engineering Mathematics

Sheffield Hallam University

Lecture 1

Learning Outcomes

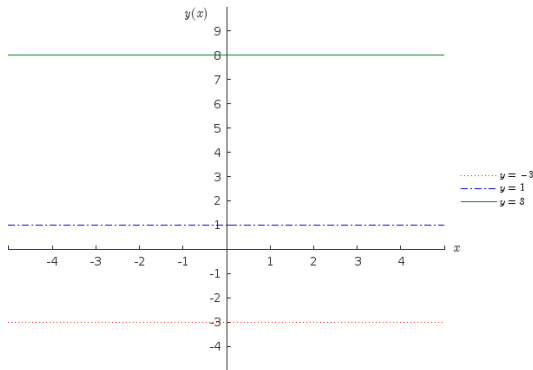
By the end of this session you should be able to:

- Recognise polynomial, exponential and logarithmic functions.
- Recognise the typical graphs of these functions.
- Determine the equation of a straight line from a graph or coordinates.
- Apply the quadratic formula.
- Use logarithms to determine unknown powers.
- Solve exponential equations.

Recognising Polynomials

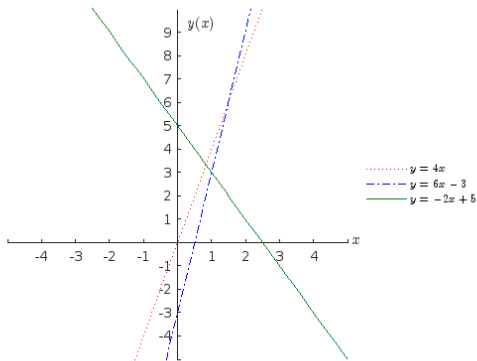
- A **polynomial function** is one that *only* involves non-negative integer powers of x , for example:
 - $y = 7x + 4$ (polynomial of order/degree 1, linear)
 - $y = 3x^2 - 5x - 1$ (polynomial of order/degree 2, quadratic)
 - $y = -x^3 + 5x^2 - 7x + 12.01$ (poly. of order/degree 3, cubic)
- Functions containing negative or non-integer powers, or other functions, (such as trigonometric) are *not* polynomials, e.g.
 - $y = x^2 + 4\sqrt{x} - 5$
 - $y = \frac{5}{x^2} - 7x^3 + 6x - 4$
 - $y = x^2 + \sin(x)$

Graphs of Polynomials: Constant functions



Graphs of **constant functions** (no dependency on x) are always straight, horizontal lines.

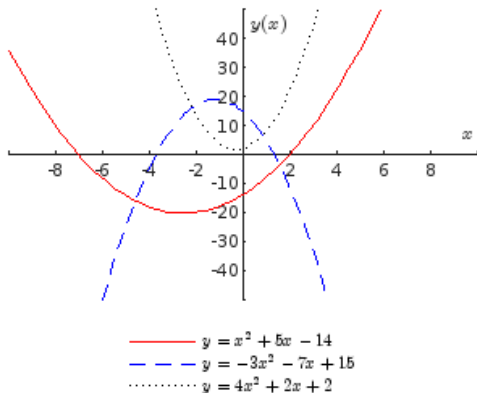
Graphs of Polynomials: Linear functions



Graphs of **linear functions** are always straight lines.

Graphs of Polynomials: Quadratic functions

Quadratic curves are either $\cup$ or $\cap$ -shaped depending on the values of a .



Determining the Linear Equation

Equation of a straight line

$$y = mx + c$$

represents a straight line, where m and c are constants.

m is the **gradient** (slope) of the line and can be calculated as

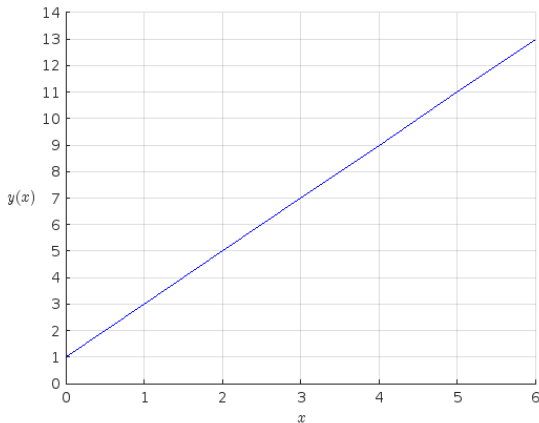
$$m = \frac{\text{vertical change}}{\text{horizontal change}} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$

c is the value of y when the line crosses the y -axis (at $x = 0$), known as the **y -intercept**.

Note: to find where the line crosses the x -axis, simply let $y = 0$.

Determining the Linear Equation

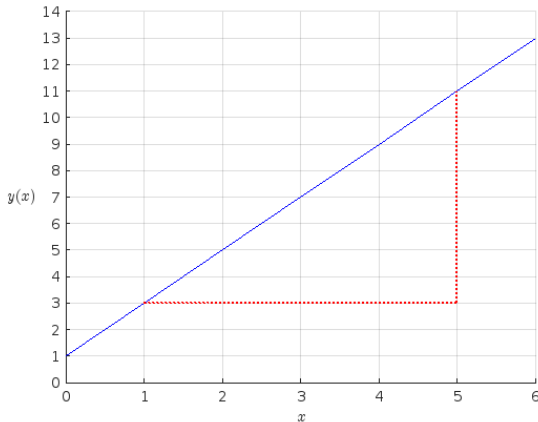
Example: Find the equation of this line:



Firstly, we can see that $c = 1$, as this is the height where the y -axis is crossed.

Determining the Linear Equation

Example: Find the equation of this line:

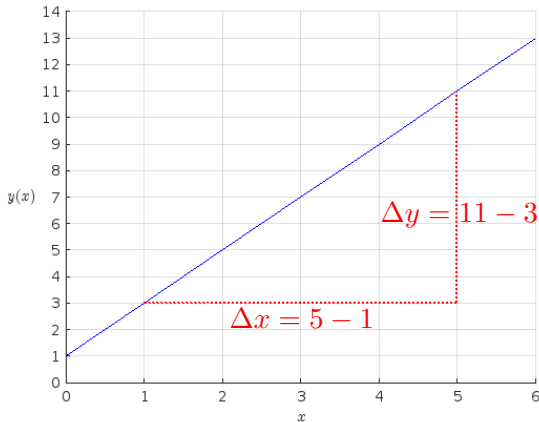


Secondly, we can determine the gradient m by constructing a triangle on the graph.

The points here are $(x_1, y_1) = (1, 3)$ and $(x_2, y_2) = (5, 11)$

Determining the Linear Equation

Example: Find the equation of this line:



$$m = \frac{\Delta y}{\Delta x} = \frac{8}{4} = 2$$

$$\therefore y = 2x + 1$$

Determining the Linear Equation

Example: Find the equation of the linear line given that it passes through the points $(1, 3)$ and $(8, -32)$.

Firstly, let's determine the gradient:

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-32 - 3}{8 - 1} \\ &= -5 \end{aligned}$$

Determining the Linear Equation

Thus we have $y = -5x + c$. To find c , we substitute one of the co-ordinates into the equation:

$$y = -5x + c$$

$$3 = -5 \times 1 + c$$

$$3 = -5 + c$$

$$\therefore c = 8$$

Therefore the equation of the linear line that passes through the points $(1, 3)$ and $(8, -32)$ is $y = -5x + 8$.

Quadratic Functions

All quadratics can be represented in the general form:

The quadratic equation

$$y = ax^2 + bx + c,$$

where a , b and c are constants and $a \neq 0$.

Quadratic Functions

The constant c is the y -intercept, as in the linear case (if $x = 0$ then the equation becomes $y = a \times 0^2 + b \times 0 + c = c$).

The curve can cross the x -axis (at $y = 0$) twice, once (just touching it) or never. If it does cross the x -axis, we can calculate the values of x where this occurs by solving:

$$ax^2 + bx + c = 0$$

The **solutions** of $0 = ax^2 + bx + c$ are the same as the x -intercepts of $y = ax^2 + bx + c$ and are also known as the **roots** of $ax^2 + bx + c$.

The Quadratic Formula

The quadratic formula:

If $ax^2 + bx + c = 0$ and $a \neq 0$ then:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad (1)$$

The **discriminant** is the “bit under the square root.” It indicates how many roots exist and of what type:

- $b^2 - 4ac > 0$ indicates two real and distinct roots (x_1 and x_2)
- $b^2 - 4ac = 0$ indicates real and repeated roots ($x_1 = x_2$)
- $b^2 - 4ac < 0$ indicates complex roots ($x = p \pm jq$)

Example

Let's use this method to solve:

$$x^2 + 4x + 3 = 0$$

In this case, the constants are $a = 1$, $b = 4$ and $c = 3$, so we substitute these into the formula:

$$\begin{aligned} x &= \frac{-4 \pm \sqrt{4^2 - 4 \times 1 \times 3}}{2 \times 1} = \frac{-4 \pm \sqrt{16 - 12}}{2} \\ &= \frac{-4 \pm \sqrt{4}}{2} = \frac{-4 \pm 2}{2} \end{aligned}$$

Thus, the solutions are $x = -3$ and $x = -1$.

Roots and Discriminants

Furthermore, if there are two distinct, real roots x_1 and x_2 to the quadratic $y = ax^2 + bx + c$, then it is possible to re-write the quadratic in the form:

$$y = a(x - x_1)(x - x_2)$$

However, if there are real and repeated roots, $x_1 = x_2$, then it is possible to re-write the quadratic in the form:

$$y = a(x - x_1)^2$$

The Logarithm Function

The logarithm function is written as follows:

Logarithm function

$$y = \log_a(x),$$

where a and x are positive and $a \neq 1$ is constant.

This can be interpreted as:

“ y is the power to which one must raise a (the base), to get x (the argument).”

That is:

$$a^y = x$$

Introduction

For example, the two statements

$$16 = 2^4 \quad \text{and} \quad \log_2(16) = 4$$

are equivalent.

Logarithms are used to simplify calculations and determine unknown powers (exponents).

Example:

What power of 2 is exactly equal to 8, i.e. $2^x = 8$?

We can write this as:

$$\begin{aligned}\log_2(2^x) &= \log_2(8) \\ x &= 3\end{aligned}$$

Examples

Calculate x :

- $3^x = 81$
- $6^x = 1$
- $2^x = 0.125$

Examples

Calculate x :

$$\bullet 3^x = 81 \quad \therefore x = \log_3(81) = 4$$

$$\bullet 6^x = 1 \quad \therefore x = \log_6(1) = 0$$

$$\bullet 2^x = 0.125 \quad \therefore x = \log_2(0.125) = -3$$

So one application of logs is to solve equations where the desired variable is in the index.

Logarithms

The most commonly used bases are 10 and e.

The **log** button on your calculator is $\log_{10}$. This is the **common** logarithm.

The **ln** button on your calculator is $\log_e$. This is the **natural logarithm** and is usually written **ln**, so:

$$\log_e(x) = \ln(x)$$

Engineers mainly deal with the natural logarithm.

Introduction - Exponential Functions

General exponential functions (with base b) are of the form:

$$y = Ab^{kx}$$

where A , b and k are constants:

- A is a coefficient.
- b is the base.
- k determines how fast the function grows (growth rate).

Certain values for the base b are more common than others, especially

$$y = 10^x \quad \text{and} \quad y = e^x = \exp(x)$$

THE Exponential Function

One particular base is very important: $e = 2.718281828$, called Euler's number.

The general form of this type of equation is:

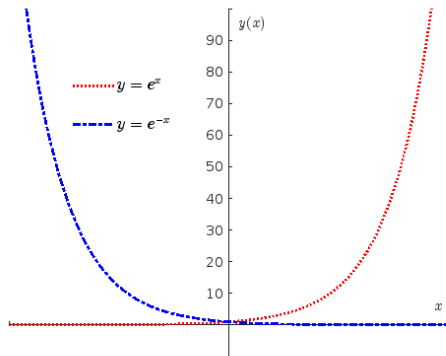
Exponential function:

$$y = Ae^{Bx} + C$$

where A , B and C are constants.

Note, when $x = 0$, $y = A + C$ (this is the y -intercept).

Visualising Exponential Functions



Changing the *sign* of B reflects the e^x curve in the y -axis: positive B gives exponential growth, negative B gives decay.

Applications of the Exponential Function

The exponential function is used frequently across engineering.

It is used in growth and decay models such as:

- Tension in belts: $T_1 = T_0 e^{\mu\theta}$
- Newton's law of cooling: $\theta = \theta_0 e^{-kt}$
- Atmospheric pressure at altitude h : $p = p_0 e^{-h/c}$
- Discharge of a capacitor: $q = Q e^{-t/CR}$

Logarithms and Exponentials

The logarithm and exponential functions are the **inverses** of each other, i.e. one undoes the impact of the other:

$$\ln(e^x) = x \quad \text{and} \quad e^{\ln(x)} = x$$

Example:

$$\ln(e^7) = 7 \quad \text{and} \quad e^{\ln(15)} = 15$$

Example 1 (I/II)

Solve the equation $4e^{-3x} + 5 = 12$ for x :

$$4e^{-3x} + 5 - 5 = 12 - 5$$

$$4e^{-3x} = 7$$

$$\frac{4e^{-3x}}{4} = \frac{7}{4}$$

$$e^{-3x} = \frac{7}{4}$$

Example 1 (II/II)

$$e^{-3x} = \frac{7}{4}$$

$$\ln(e^{-3x}) = \ln\left(\frac{7}{4}\right)$$

$$-3x = \ln\left(\frac{7}{4}\right) \quad [\text{now divide by } -3]$$

$$x = -0.187 \text{ to 3 d.p.}$$

Example 2

A capacitor of capacitance C is allowed to discharge through a resistor of resistance R such that the voltage across the terminals of the capacitor v at time t after the discharge started is given by:

$$v = v_0 e^{\left(-\frac{1}{RC}t\right)},$$

where v_0 is the voltage across the terminals of the capacitor at the start of the discharge.

If $C = 500 \text{ nF}$, $R = 200 \text{ k}\Omega$ and $v_0 = 12 \text{ V}$, determine the time it takes for v to drop to 6 V .

Example 2 - Solution

Substitute the values:

$$6 = 12e^{\left(-\frac{t}{200 \times 10^3 \times 500 \times 10^{-9}}\right)}$$

Simplifying:

$$\frac{6}{12} = e^{\left(\frac{-t}{10^{-1}}\right)} \quad \implies \quad \frac{1}{2} = e^{-10t}$$

Using $\ln$ to invert the exponential:

$$\ln\left(\frac{1}{2}\right) = \ln\left(e^{-10t}\right) \quad \implies \quad t = -\frac{1}{10} \ln\left(\frac{1}{2}\right) = 0.069 \dots$$